Agentic Crystallization v2.4

Independent Verification, Adaptive Trust, and Mathematical Cooling

Agentic Crystallization Research

October 2026

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Figure 1: Agentic Crystallization

Agentic Crystallization v2.4 — Independent Audit, Adaptive Trust, and Cooling to a Minimal Rule

Research whitepaper · Synthetic evaluations · Historical continuity v1.0–v2.3 preserved

Abstract

Agentic Crystallization investigates whether an adaptive system can discover, validate and retain useful interacting procedures under cost and uncertainty constraints. Versions 2.0–2.3 implemented a bounded synthetic adaptation loop, renewed certificates under environmental change, tested risk-based verification and showed that two feedback streams can agree while jointly misleading an agent. Version **2.4** adds an independently sampled third audit reference, optional risk-triggered rechecks, counters for gross channel disagreement, automatic quarantine on contradiction, and an explicit reference-compromise falsification test.

The principal finding is **conditional**: with an honest reference, the triple-check controller records zero incorrect *fresh* certifications when primary and anchor are jointly corrupted, compared with 402 for the v2.3 two-channel guard (80-world holdout). Its mean synthetic utility in that regime is +3.16 per epoch, compared with -11.31 for the guard. Conversely, in honest stationary, gradual, or recurring environments the extra audits reduce utility. If all three channels share a corruption, v2.4 still issues 320 incorrect fresh certifications. There is no universal trust inference from mere agreement.

The development search selected a four-epoch audit interval with cue-triggered checks disabled. **Its held-out actions exactly matched a simpler four-epoch periodic controller**, a negative result against unnecessary complexity. The closed loop remains a synthetic research prototype, not a proven self-sustaining computational crystal.

1. Scientific objective and continuity

The research began with an analogy between physical crystallization and the selection of reusable configurations of procedures. That analogy does not identify a physical law. Historical mechanisms include Möbius higher-order interactions (v1.0–1.1), conditional transfer and negative transfer (v1.2–1.3), Bayesian information acquisition (v1.4–1.6), robust decision certificates (v1.7–1.8), sequential evidence efficiency (v1.9), persistent candidate-selection loops (v2.0), environmental renewal (v2.1), risk-adaptive verification (v2.2), and dual-channel integrity limits (v2.3).

The new falsifiable question is narrower: **How much additional independent feedback is needed to protect against channel corruption, and is its value worth the cost?** The full historical source, datasets and reports remain in this archive. Previous v2.3 documentation is separately archived.

2. Mathematical object and conditional theorem

At epoch t, a bounded synthetic gain $g_t(a)$ describes either procedure A or B relative to a zero-gain fallback. The policy does not know g_t. Each measurement channel supplies a bounded sample average; the reference's random seed is independent of the original two streams when it is honest.

With $T = 24$ epochs, $K = 2$ actions, $C = 3$ channels and familywise error budget $\alpha = .05$, define:

$$r_c(n) = \sqrt{\frac{2\log(2TKC/\alpha)}{n}}.$$

If each reference observation is independent, bounded and has expectation equal to the *current* action gain, then the reference-only lower/upper interval simultaneously covers those gains across the pre-specified horizon with probability at least $1-\alpha$. A fresh certificate is issued only if the reference's lower bound exceeds zero and the competing action's upper bound, and the other two channels separately support the same action without significant discrepancy.

This guarantee is conditional on honest reference measurements. It is not an empirical proof of honesty. Stale deployments in intervening epochs are not covered. Details and proofs are in Mathematics v2.4.

3. The research algorithm

The v2.4 controller implements **pre-committed audits**: before the independent reference result can enter the policy, a scheduling function decides whether to pay for it. Then the controller either certifies a current procedure, retains a previously supported choice with an explicit stale label, or selects the safe baseline. Significant discrepancies quarantine the evidence and trigger bounded follow-up checks rather than a majority vote.

The measured disagreement indicator is $D = 1$ if any corresponding sample means differ by more than their simultaneous error radii. A Beta(1,1)-smoothed mismatch frequency is retained as an **uncalibrated descriptive statistic**, not a probability of truth. No version of the controller is permitted to infer ground truth solely from multiple channels agreeing.

Development compared eight predeclared parameter configurations: maximum verification ages of 3, 4, 6 and 8 epochs, each with cues enabled or disabled. The chosen setting was **four epochs without cue-triggered audits**. This matches simple scheduled verification on the independent holdout; therefore the complex scheduler is not justified as a production default.

4. Experimental design

- **Simulator:** 24 epochs per world, two whitelisted candidate procedures and zero-gain baseline; true gains defined by a planted task generator.
- **Observations:** 384 primary and 384 anchor samples per action, 1,024 independent reference samples per action when purchased; each channel can be separately corrupted.
- **Costs:** primary .002, anchor .006, independent reference .006 synthetic units per observation; task reward scaled to 80 task instances per epoch.
- **Policies (8):** tuned triad adaptive, simple periodic triad, every-epoch triad, reference-only periodic, reference-only every-epoch, v2.3 dual guard, primary-only periodic, and zero-gain baseline.
- **Development:** 18 seeds (4200–4217), nine comparison regimes; hyperparameters frozen before holdout.
- **Holdout:** 80 fresh seeds (9000–9079), 13 regimes, 1,040 worlds, 199,680 policy-epoch records.
- **Independent replication:** 40 seeds (11000–11039), 520 worlds, 99,840 records.
- **Controls:** honest stationary and shifting tasks; one-channel corruption; two-channel shared corruption; compromised reference; all-channel shared corruption; missing reference or anchor; misleading warning; unlearnable tasks.
- **Integrity:** code-source hashes frozen before holdout, deterministic recorded seeds, SHA256 manifests, read-only receipt verifier and clean independent replay.

5. Full held-out outcome table

Mean synthetic net utility per epoch, higher is better. The last two columns count v2.4 triad incorrect fresh certificates and discrepancy alarms across all 80 worlds, 24 epochs each. They do not count stale harmful deployments.

Regime	v2.4 triad	v2.3 guard	Primary only	Reference only	Triad false fresh	Triad alarms
Stationary	+23.37	+25.99	+27.62	+24.93	0	0
Abrupt change	+23.29	+25.82	+27.56	+24.93	0	0
Gradual drift	+9.66	+14.69	+17.41	+16.01	0	0
Recurring patterns	+13.84	+24.09	+18.22	+15.59	0	0

Regime	v2.4 triad	v2.3 guard	Primary only	Reference only	Triad false fresh	Triad alarms
Corrupted primary	+3.16	+4.49	-9.72	+24.93	0	479
Corrupted anchor	+3.19	+4.49	+27.62	+24.93	0	480
Shared primary + anchor error	+3.16	-11.31	-9.66	+24.93	0	479
Compromised reference	+3.16	+25.82	+27.56	-12.41	0	479
All three compro-mised	-13.96	-11.31	-9.66	-12.41	320	0
Reference missing	+7.76	+25.82	+27.56	+8.31	0	0
Anchor missing	+4.31	+7.49	+27.56	+24.93	0	0
Misleading warning	+23.39	+20.04	+27.62	+24.93	0	0
No learnable advantage	-6.14	-1.89	-0.38	-4.10	0	0

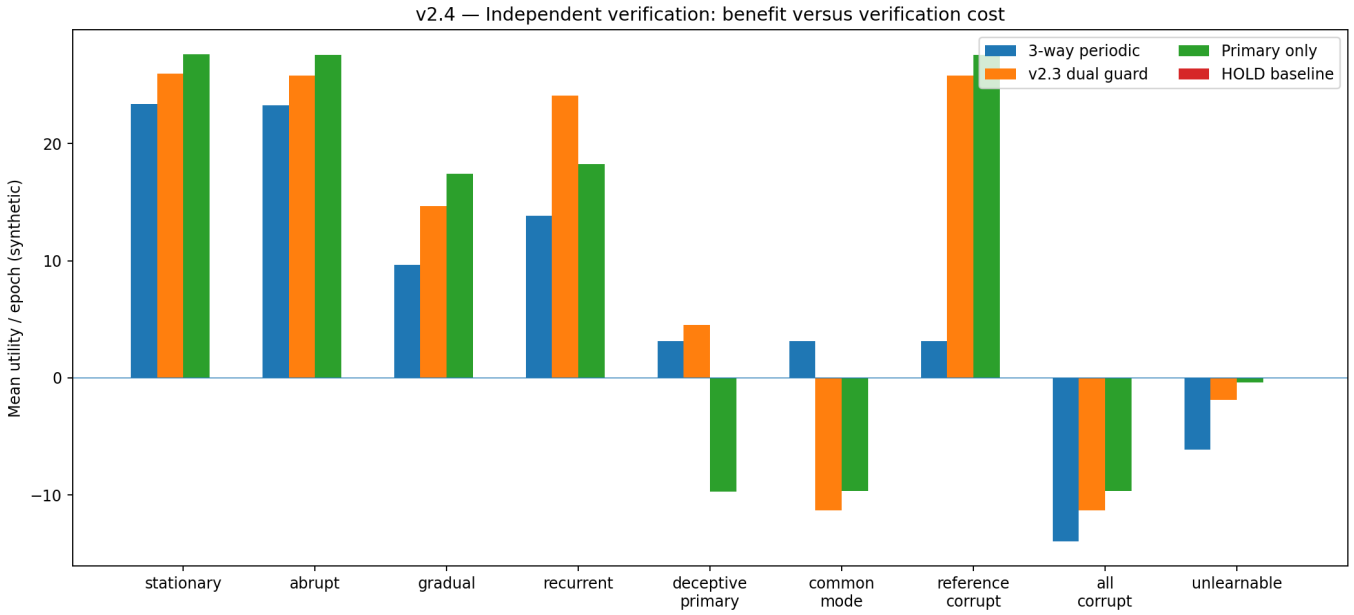


Figure 2: Net synthetic utility across environments

Focused comparisons

- **Honest stationary:** triad +23.37, guard +25.99, primary-only +27.62. Extra verification costs utility.
- **Gradual drift:** triad +9.66, guard +14.69. The triad's extra checks do not solve changing latent task structure.
- **Recurring patterns:** triad +13.84, guard +24.09. Independent reference can over-verify a cheap memory reuse process.

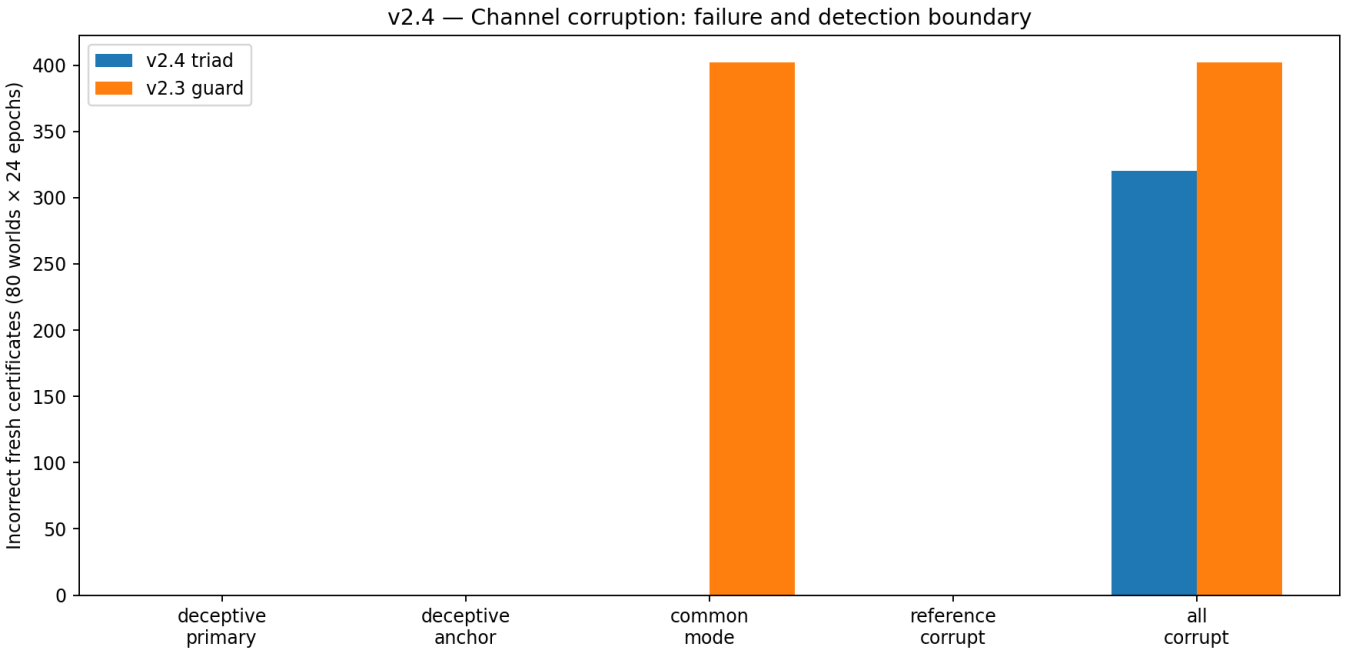


Figure 3: Incorrect fresh certificates under corruption

- **Common-mode corruption of the original pair:** triad +3.16 versus guard -11.31; triad 0 incorrect fresh certifications and 479 alarms versus guard 402 incorrect fresh certifications and no alarms.
- **Reference compromised:** triad +3.16 versus guard +25.82. Disagreement allows safe quarantine but not attribution of which channel is wrong.
- **All channels compromised:** triad -13.96, 320 incorrect fresh certifications and 0 alarms. More agreeing validators cannot identify systematically wrong evidence.
- **Unlearnable environment:** triad -6.14, baseline 0; spending audit resources without a learnable advantage remains a failure.

These are effects produced by this exact synthetic simulator, not predictions of Pegasus behavior. Independent replication reproduced the qualitative results, including 0 triad fresh false certifications under two-channel corruption with an honest reference and 160 incorrect fresh certifications when all three channels were corrupted.

6. Statistical and epistemic limits

The Hoeffding theorem bounds false **fresh** certifications across fixed epochs under honest independent reference observations. It cannot certify audit provenance, prevent changes after a measured epoch, guarantee outcomes of stale deployments, or detect correlated corruption of all observables. Adding sensors only creates an external cross-check if those sensors really provide independently grounded evidence; otherwise, the indistinguishability impossibility result applies.

Training-seed performance was used to choose the audit policy. Holdout bootstrap contrasts are descriptive and not claims of preregistered confirmatory significance. Neither tuning nor the observed success justifies silently increasing the policy's real-world authority.

7. Algorithmic consolidation result

A stable research design is allowed to get simpler. v2.4 shows that a richer adaptive audit schedule can collapse to a fixed schedule on independent data. Retaining the sophisticated heuristic without independent evidence of benefit would violate the research objective of evidence-governed optimization.

We therefore **keep three-way periodic verification as the experimental integrity safeguard**, keep the v2.3 two-channel guard as a lower-cost alternative under justified assumptions, and preserve the full v2.4 adaptive implementation solely for ablation and future hypotheses.

8. Next falsifiable research question

The unsolved problem is *not* to manufacture higher confidence by accumulating correlated signals. It is to establish a trustworthy audit provenance and sample an independent consequence of deployed decisions under real tasks. A future experiment should test independently adjudicated outcome labels, intervention cost, audit compromise, slow adversarial corruption below detection thresholds, and budgeted abstention. Without genuinely identifying observations, the task is provably impossible for some pairs of environments.

No claim is made that digital computation has undergone a physical crystallization phase transition, that v2.4 produces indefinite self-improvement, or that Pegasus/Hermes was executed or improved.

9. Reproducibility and evidence links

Run instructions · Source · Stage P driver · Raw holdout · Protocol · Development tuning · Frozen code · Replication · Claim ledger · Historical prior.

Agentic Crystallization v2.4 — Formal Mathematics

Status: conditional finite-sample guarantees and falsifiable impossibility results; all observed outcomes are synthetic.

1. The problem and evidence channels

Let time be indexed by epoch $t \in \{0, \dots, T-1\}$ with $T = 24$, candidate actions $a \in \{A, B\}$, and a fallback a_0 with known synthetic gain zero. Let the true current-epoch gain relative to that fallback be $g_t(a) \in [-1, 1]$. The algorithm does not observe $g_t(a)$ directly. For each channel $c \in \{P, A, R\}$ (primary, anchor, independently generated audit reference), it can **purchase** a sample mean

$$\hat{\mu}_{t,a,c} = \frac{1}{n_c} \sum_{i=1}^{n_c} X_{t,a,c,i}, \quad X_{t,a,c,i} \in [-1, 1].$$

Honest channel assumption: the observations within an action/epoch/channel are independent and their common expectation is $g_t(a)$. The reference is simulated with independent random draws conditional on the latent truth, but **independent randomization does not itself establish the validity of the measurement**. In corruption experiments the assumption is deliberately false. All inputs to the actual decision are sampled means, a cue, time and retained state; the true gain is reserved for scoring.

2. The finite-horizon simultaneous certificate

Theorem 1 (honest reference, familywise coverage). Prespecify T epochs, K candidate actions, C candidate measurement channels, and confidence budget $\alpha \in (0, 1)$. If each selected reference sample mean is composed of n_R independent bounded observations in $[-1, 1]$ whose expectation is the *current* $g_t(a)$, then setting

$$r_c = \sqrt{\frac{2}{n_c} \log\Big(\frac{2TKC}{\alpha}\Big)}$$

ensures simultaneous coverage of all **honest** channel means across this finite catalog and time horizon with probability at least $1-\alpha$. Conditional selection or stopping does not invalidate the result, because the union bound covers all prespecified candidate means.

Proof. Hoeffding's inequality for an average of independent variables in $[-1, 1]$ implies $P(|\hat{\mu} - \mu| > r) \le 2e^{-nr^2/2}$. Set the right-hand side to $\alpha/(TKC)$ and sum over the finite prespecified collection. If only the reference channel is honest, its part of the union bound alone still guarantees simultaneous reference coverage with probability at least $1 - \alpha$ (indeed more conservatively). □

Here $T = 24$, $K = 2$, $C = 3$, $\alpha = .05$, $n_P = n_A = 384$, $n_R = 1024$. Each comparison with the fallback is with a baseline known to be exactly zero **inside the synthetic model**.

Corollary 1 (honest-reference fresh certification). Define reference-favored candidate $a^* = \arg\max_a \hat{\mu}_{t,a,R}$. If

$$\boxed{\hat{\mu}_{t,a^*,R} - r_R > \max\big(0, \max_{b \neq a^*}(\hat{\mu}_{t,b,R} + r_R)\big)},$$

then on the simultaneous reference-coverage event $g_t(a^*) > 0$ and $g_t(a^*) > g_t(b)$ for all competitors b. Requiring corroboration by primary and anchor channels may reject more candidates; it **cannot invalidate this conclusion** while the reference remains honest. A certificate is for the *current epoch only* and is not a warranty of future performance.

3. Three-way discrepancy rule

Two channel means are statistically incompatible if

$$D_{cR,t} = \mathbf{1}\left\{\exists a : |\widehat{\mu}_{t,a,c} - \widehat{\mu}_{t,a,R}| > r_c + r_R\right\}.$$

If both channels are honest measurements of the same current gain, then $D_{cR,t} = 0$ on the simultaneous coverage event. A positive alarm rejects their joint-honesty assumptions; it **does not identify which channel is untrustworthy**. When either channel disagrees, v2.4 chooses HOLD and records the event. Mere statistical compatibility does not imply either one's honesty.

An accepted fresh certificate additionally requires matching **separately certified best actions** across all three channels. This requirement lowers false promotions when the reference is honest but can cause false negatives and severe utility loss when one channel is compromised or absent. It is conservative corroboration, not majority voting.

4. Learned-looking trust scores are not safety guarantees

After m observed comparisons and f gross discrepancies, the code reports

$$\tilde{p}_{\text{mismatch}} = \frac{1+f}{2+m}.$$

This is the mean of a Beta$(1+f, 1+m-f)$ distribution **only under a Bernoulli exchangeability model**. Here, audits are selected from time-varying regimes, so neither exchangeability nor independent reliability labels are assured. Accordingly this number is an **unverified descriptive diagnostic**, never the premise of a certification claim. The only mathematical certification depends on the externally stated honesty assumption and finite-sample concentration.

5. A cost-aware audit decision

Let C_R denote the charged reference+primary+anchor audit cost and let h be declared hazard, ℓ_t the last *observed* estimated gain, and d_t the number of epochs since the last audit. The experimental adaptive trigger is

$$\boxed{\min(1, hd_t)\, N \max(0, \ell_t) > C_R,}$$

or expiration at maximum age, or an enabled warning cue. No true outcome or future shift label appears in this trigger. The units are synthetic utility points. This is a heuristic **not** a calibrated posterior predictive value-of-information theorem, since the hazard and future harm model are not proven.

v2.4 tuned $d_{\max} \in \{3, 4, 6, 8\}$ and cue enablement on disjoint development worlds. The selected combination was **four epochs, no cue**. On the final holdout, its actions and outcomes coincided with simple four-epoch periodic three-way verification. **The extra scheduling complexity did not demonstrate value and should not be promoted as necessary.**

6. Indistinguishability: more agreeing sensors cannot prove truth

Proposition 2 (common-mode observational equivalence). Consider two worlds with exactly the same joint distribution over all observed channel outputs, warning cues and retained histories, but opposite true gains for the two candidate actions. No policy measurable with respect to the observed history can guarantee a non-baseline optimal choice in both worlds.

Proof. Under equal observation laws, any deterministic or randomized policy has the same conditional distribution over actions in both worlds. Since the optimal actions differ by construction, a non-baseline choice that succeeds in one must fail in the other. Randomization cannot eliminate this

contradiction. Abstention can avoid falsely asserting which unobserved world is real, but provides no universal improvement guarantee. □

An externally audited reference can separate the worlds **only if it adds a valid observation whose outcome law differs across them**. If every channel is compromised in the same way, adding their agreements changes nothing about identifiability.

7. Safety, costs, and temporal limits

All deployments after the audit epoch and before renewed evidence are flagged **stale**, even if recently certified. Fresh-certification validity does not extend across unknown shift, interference, nonstationarity, delayed feedback or adversarial adaptation. The fallback is known zero-gain *in this specific toy environment*. Audit cost and task reward are measured in the same synthetic units:

$$J(\pi) = \frac{1}{MT} \sum_{w,t} (N g_{w,t}(a_{\pi,t}) - C_{\pi,t}^{\text{audit}}).$$

The controller also reports negative deployments, incorrect fresh certifications, refusal/HOLD counts, and alarms. A policy that rejects everything avoids incorrect fresh certifications but may be economically useless; therefore the utility and coverage metrics must always be shown alongside false-cert counts.

8. Claim ladder and falsification

1. **Exact mathematics:** conditional simultaneous finite-horizon Hoeffding certificates, discrepancy implications, and the indistinguishability proposition.
2. **Synthetic empirical observation:** three-way periodic verification catches shared corruption of the first two channels under an honest reference but is expensive and not universally superior.
3. **Negative result:** an added adaptive scheduling heuristic tied simple periodic verification in all 1,040 holdout worlds; it is **not** an established improvement.
4. **Open:** how to validate independent audit provenance, diagnose shared corruption without trusted interventions, and optimize verification economics under real-world tasks.
5. **Not demonstrated:** general self-sustaining improvement, novel physics, critical phase transition, Pegasus or Hermes performance gains.

Whitepaper · Algorithm card · Held-out protocol · Complete source

Agentic Crystallization v2.4 — Algorithm Card

Object: audited two-candidate bounded synthetic procedure selector. No Pegasus integration or promotion authority.

Inputs

`policy`, `epoch`, `warning`, retained `TrustState`, observed primary/anchor means and *only if purchased* reference means, `TrustConfig`.

State

Selected action or baseline, last reference audit epoch, disagreement/quarantine status, capped exponential retry epoch, last observed gain, per-channel discrepancy counts, separate historical v2.3 guard state.

Outputs

Selected action, whether a **fresh** current-epoch certificate was issued, stale deployment indicator, HOLD, channel-disagreement alarms, paid audit counters, net audit cost, and retained state.

Two-phase procedure

1. `request_reference()` commits to a paid independent reference audit from prior state, time, cue and costs, **before** seeing the new reference result. If no audit is purchased, the decision function receives `None` for the reference.
2. On audit, obtain reference outcomes and, for triad policies, primary and anchor outcomes. If either mandatory channel is unavailable, return HOLD.
3. Compute simultaneous Hoeffding radii for every sample source, action, epoch. If primary/reference or anchor/reference intervals are disjoint, HOLD and quarantine. Neither disagreement nor agreement identifies the honest source.
4. Issue **fresh** certificate only if every channel independently certifies the same action over the fallback and competing action, with no gross discrepancy. On failure, HOLD and back off costly rechecks at a bounded rate.
5. Between audits, retain decisions only while age is bounded, mark them as **stale** and never call them fresh. On expiry or contradiction, HOLD.
6. Charge actual audit costs and record all outcomes. The synthetic hidden true gain is accessed **only after policy decision**, for evaluation.

Selected research policy

Development-only grid search chose `max_age=4`, `use_cue=False`. In the independent holdout this adaptive policy exactly matched simple `triad_periodic` with four-epoch intervals. **Prefer the simpler implementation unless a future independent experiment proves the heuristic helps.**

Correctness premises

Prespecified finite horizon and candidate catalog, bounded independent outcomes within the honest reference, exact current gain expectation, no post-hoc use of ground truth or holdout tuning. Under these premises, current-epoch fresh miscertification risk is controlled by the simultaneous concentration bound. Stale reuse has no such guarantee.

Documented failure cases

Shared corruption of all channels → confident incorrect certification. Reference compromised → HOLD costs and lost task utility. Missing channel → HOLD. Unlearnable tasks → excess audit costs. Unanticipated drift between audits → stale harm. Selected audits and nonstationarity make posterior mismatch rates *not* automatically calibrated.

Formal proof · Full whitepaper · Executable tests · Source